
WORK EXPERIENCE

- **Cisco** Vancouver, BC
Software Engineer G10 May 2020 – Present
 - Developed cloud networking service that enables service-to-service, datacenter-to-datacenter connectivity for **100k+** users at 99.999% uptime via a software-based multi-tenant router
 - Designed and built a self-service web tool that significantly reduced time to customer incident resolution. It enabled real-time inspection of the customer network configuration, packet path tracing, and in-browser packet capture
 - Led **FedRAMP certification** for service, coordinated between teams to meet stringent US federal requirements
 - Reduced customer downtime risk by halving deployment and failover times through optimizing GoBGP internals to efficiently process 1M+ route updates.
 - Maintained in **C** custom dataplane engine that facilitates GENEVE/VxLAN overlay for both IPv4/IPv6 and MPLS-like capabilities
 - Helped drive major infrastructure-as-code (IaC) and application-level changes to ensure service was multicloud ready (AWS, private cloud, OCI, etc.)
 - Significantly improved operational costs and resilience by implementing cross-datacenter discovery and failover of downstream services, accounting for geolocation and customer presence to minimize latency. Resulted in US Patent 12081430 for this work
 - Developed containerized proxy client that enabled secure connection from the customer's private cloud to Cisco's cloud for easy Zero Trust Access
 - Implemented MQTT-based integration with AWS' IOT to enable real-time telemetry from proxies; and ACME integration to manage certificate lifecycle management
 - Maintained the full observability stack (OpenTelemetry, Prometheus, Grafana, Datadog, Splunk), including VM metrics exporters and Terraform-driven monitors/dashboards, ensuring platform-wide visibility and alerting
 - Resolved several high-severity customer issues, participated in on-call rotation, led postmortems, and drove follow-up actions that improved alerting precision and resolution times
- **Qumulo** Vancouver, BC
Software Engineer Intern May – Aug 2019
 - **Qumulo Distributed File System:** Helped implement in **C** data Snapshot Replication and retention across distributed file systems, working both back-end to interface with file system APIs and creating **Typescript** Web UIs for convenient usage
- **Intel of Canada, Ltd.** Vancouver, BC
Firmware Engineer Intern May – Dec 2018
 - **SSD Media PHY Driver:** Assisted engineers validate new product boards by rapidly developing firmware to initialize and validate PHY drivers to NAND/DRAM memory, such as margins training and dynamic PLL changing, enabling them to analyze waveforms and verify signal integrity
- **OSI Maritime Systems Intern** Burnaby, BC
Software Engineer Jan – Apr 2017
 - **Arctic Offshore Patrol Ship Project:** Assisted developers in meeting tight project deadlines by developing **Bash** scripts that could interface with multiple machines and asynchronously deploy latest company builds so that any last-minute tweaks could be tested on the production floor

SKILLS

- Go, Python, Bash, C, Vue
- Docker, Ansible, Terraform, AWS, Openstack, OCI, Linux, OpenTelemetry, Prometheus, Thanos, Grafana, Splunk, Datadog
- BGP, DNS, DHCP, HTTP/S, TCP/IP, UDP, IPsec, ICMP, MPLS

EDUCATION

- **University of British Columbia** 2015 – 2021
BaSc.in Engineering Physics